

NAVUUNA · BUILD SPEC · 21 AUGUST 2026

Refocus workflow & Claude Code prompts

Everything the team does next: the data pipeline, the software sequence, seven scoped prompts, and the institutional track that runs alongside.

REPO

github.com/AustineIgunza/Nuvola_Atlas

READS WITH

Focus & Business Value Brief
One-Page Positioning
Data Sourcing Findings Report, 16 Aug 2026

SECTION 0

What actually changes

Nothing in the stack is thrown away. Laravel 11, FastAPI, Supabase/PostGIS, React 18 + Vite 5 and the Mapbox map all stay exactly as they are. What changes is what feeds them.

	BEFORE	AFTER
Pillars rendered	8 — four real, two proxy, two with no source at all	3 — two measured, one labelled proxy
Flagship	"Vitality Score"	Water & Sanitation service performance
Second	Road Progress	Road density + transit accessibility
Held, labelled	Smart Grid	Electricity Access (2019 census)
Switched off	—	Freedom Index · Safety · Project Momentum · Environment
New asset	—	WASREB IMPACT time series — 17 editions, machine-readable
New requirement	—	Provenance on every value — source, vintage, granularity, method

Same engine. Less fuel, better fuel, with the receipts attached.

SECTION 1

The one decision that will trip you up

Granularity mismatch. Settle this in week one or it poisons everything downstream.

- KNBS census data is **sub-county** — 17 units. It maps cleanly onto your bubbles.
- WASREB IMPACT data is **utility-level**. NCWSC serves the whole of Nairobi County.

So Nairobi's 51% non-revenue water is **one number covering all 17 sub-counties**. You cannot spread it across bubbles. Doing so would be inventing data — exactly what your own integrity rule forbids.

How to handle it

Every indicator carries a granularity field: subcounty | county | utility | national.

- subcounty → renders on the bubble
- county / utility → renders in a **county banner above the map**, never on bubbles
- national → context only, greyed
- Anything with method: "gap" → grey, no number, ever

READ THIS BEFORE ANYONE WRITES THE PARSER

This is not a limitation to hide. It is the product. Nobody else in Kenya publishes a service-performance figure with its granularity honestly declared — which is exactly why a regulator or funder will trust yours.

SECTION 2

Three parallel tracks

TRACK A · DATA

Produce the files. Runs offline. No backend needed.

Joy leads, Austine supports

TRACK B · SOFTWARE

Security first, then scope cut, then consume the files.

Devyan + Khillon

TRACK C · INSTITUTIONAL

Letters and registrations. Blocks nothing; unblocks everything later.

Austine

A and B are decoupled by ONE contract: the field names inside `nairobi_vitality.geojson`

Agree them on day one and neither track ever blocks the other.

FIG. A — The three tracks and the single contract that decouples them.

SECTION 3

Data workflow — Track A

Stage 0 — Boundaries (do this first; everything joins to it)

ITEM	DETAIL
Source	Kenya COD-AB admin2 (IEBC), hosted on the Humanitarian Data Exchange
URL	data.humdata.org/dataset/cod-ab-ken
Take	the admin2 shapefile from inside the .shp.zip
Licence	Open — attribute IEBC / geoBoundaries

```
ogr2ogr -where "adm1_name = 'Nairobi'" nairobi_subcounties.gpkg ken_admin2.shp
ogrinfo -so nairobi_subcounties.gpkg      # must report 17 features
```

If it reports 0, run `ogrinfo -so ken_admin2.shp` and find the real county-name column.

Stage 1 — Sources, by pillar

Water & Sanitation — flagship

SOURCE	WHAT	GRANULARITY	EFFORT
KNBS 2019 Census	% households by main water source; by toilet type	sub-county	Low — it's an xlsx
WASREB IMPACT 1–17	NRW, hours of supply, metering ratio, revenue collection, O&M cost coverage, water quality, staff productivity, coverage	utility + county	High — this is the build
WPDx	Individual water points with status	point	Medium

Road — second

SOURCE	WHAT	GRANULARITY	EFFORT
HOT OSM roads (Kenya)	Road network as lines	line → aggregate	Low
Digital Matatus GTFS	Matatu routes and stops	point / line	Medium

Compute two things: road density (km per km²) and **transit accessibility** — share of population within 500 m of a matatu stop, per sub-county. The second is the interesting number, and nobody publishes it.

Electricity Access — held, labelled

KNBS 2019 Census: % of households using electricity for lighting. Sub-county. Label the vintage loudly, everywhere it appears.

Stage 2 — The WASREB parser (the crown jewel)

The single highest-value artefact in the whole project. Treat it accordingly.

Input: 17 annual IMPACT report PDFs from wasreb.go.ke/impact-reports
Output: `wasreb_impact_long.csv`
Schema: `utility_id, utility_name, county, fy, indicator, value, unit, report_issue, page_ref, extraction_confidence`

Four notes that will save you a week

- The indicator set and table layout **change between editions**. Do not write one parser — write per-edition extractors sharing a normaliser and a canonical indicator vocabulary.
- Keep `page_ref` on every value. When WASREB asks how you got a number, you show them the page.
- `extraction_confidence`: high (clean table), medium (OCR/reflow), low (ambiguous). Publish the low ones flagged — never silently drop them.
- Spot-check **at least 30 values by hand** against the PDFs before publishing. One wrong figure and the entire record is worthless.

Stages 3 & 4 — Emit and publish

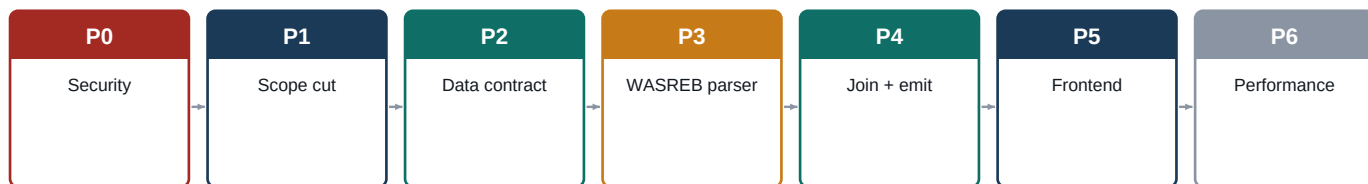
Three outputs, not one:

- `nairobi_vitality.geojson` — feeds the map, in exactly the shape the Integration Guide specifies
- `wasreb_impact_long.csv` + `.parquet` — the open asset you publish
- `provenance.json` — source, licence, vintage, retrieval date and checksum per dataset

Static file in the repo, per Option A in the Integration Guide. No backend until there is a genuine live-refresh requirement. Then publish the WASREB series openly, attributed to WASREB.

SECTION 4

Software workflow — Track B



P0 BLOCKS EVERYTHING ELSE. Run each as a separate Claude Code session and review before moving on.

FIG. B — The order is not negotiable.

SECTION 5

Repo layout after the refocus

Add a fourth package to the monorepo.

```

Nuvola_Atlas/
|-- nuvola-atlas-backend/      (Laravel 11)
|-- nuvola-atlas-frontend/    (React 18 + Vite 5)
|-- nuvola-atlas-ingestion/   (FastAPI)
`-- nuvola-atlas-data/        <-- NEW
    |-- manifests/            one YAML per source: url, licence, vintage, checksum
    |-- sources/              raw downloads (gitignored; manifests are the record)
    |-- pipeline/
    |   |-- boundaries.py
    |   |-- knbs.py
    |   |-- wasreb/           per-edition extractors + normaliser
    |   |-- osm_roads.py
    |   |-- gtfs_transit.py
    |   |-- indicators.py    canonical vocabulary + granularity rules
    |   |-- score.py         scaling + weights, in ONE place
    |   `-- emit.py
    |-- outputs/
    |   |-- nairobi_vitality.geojson
    |   |-- wasreb_impact_long.csv
    |   `-- provenance.json
    `-- tests/
  
```

Keep the scoring weights in one versioned file. An index whose weighting is hidden cannot be defended in a room.

SECTION 6

CLAUDE.md for the repo root

Drop this at the repo root. Claude Code reads it automatically on every session, which stops you re-explaining the scope each time — and stops the scope quietly creeping back.

```
# Navuuna (repo: Nuvola_Atlas)
```

```
Sub-county service-performance record for Nairobi. NOT a general urban intelligence platform. Scope was deliberately narrowed in Aug 2026.
```

```
## Active pillars
```

```
- Water & Sanitation (flagship) -- KNBS census + WASREB IMPACT
- Road (second) -- OSM road density + Digital Matatus GTFS
- Electricity Access (held) -- KNBS 2019 census, labelled with vintage
```

```
## Switched off -- do not reintroduce without a written decision
Freedom Index . Safety & Security . Project Momentum . Environment
Smart Grid (live)
```

```
## Non-negotiable rules
```

1. Never render an invented or proxy number as measured.
method: "gap" renders grey with no number.
2. Every indicator value carries source, vintage, granularity, method.
3. Utility/county-level values NEVER render on sub-county bubbles.
County banner only.
4. The AI assistant has no access to `users` or any table containing PII.
5. Scoring weights live in one versioned file. No inline magic numbers.

```
## Stack
```

```
Laravel 11 . FastAPI (Python 3.13) . Supabase Postgres + PostGIS
```

```
React 18 + Vite 5
```

```
Packages: nuvola-atlas-backend, -frontend, -ingestion, -data
```

SECTION 7

The seven Claude Code prompts

Run these as separate sessions. Do not paste them all at once — Claude Code does better work on scoped tasks with clear acceptance criteria, and you want to review each one before starting the next.

P0 — Security remediation · RUN FIRST

Context: Laravel 11 API with an LLM-backed chat feature that generates SQL against a Supabase Postgres database. A security audit found critical issues. This code will handle Kenyan government data, so this must be airtight.

Findings to fix:

1. CRITICAL - `users` is listed in `allowed\_tables` in config/ai.php, and a migration grants SELECT on public.users to the read-only chat role. A prompt injection can therefore retrieve emails and password hashes.
2. CRITICAL - In local/testing environments without DB_CHAT_RO_USER configured, execution falls back to the primary DB_USERNAME connection, which has full read/write privileges.
3. HIGH - SqlGuard validates table names with the regex `/\b(?:FROM|JOIN)\s+([A-Z0-9_.]+)/` which misses implicit comma joins. ``SELECT * FROM zones, users`` passes the guard.

Tasks:

- Remove `users` from allowed_tables.
- Write a migration that REVOKES SELECT on public.users from the chat role and creates a view `chat\_user\_stats` exposing only id, role, created_at. Add only that view to the allowlist.
- Make DB_CHAT_RO_USER mandatory in ALL environments. If it is not configured, the chat feature must fail closed with a clear error - never fall back to the privileged connection. Update .env.example and the local setup docs.
- Replace the regex table extraction with a proper SQL AST parser (evaluate greenlion/php-sql-parser or an equivalent maintained library). Validate every table reference node against the allowlist, including comma joins, subqueries and CTEs.
- Add tests covering: comma-join bypass, subquery reference to a disallowed table, CTE reference, semicolon inside a string literal, and a missing DB_CHAT_RO_USER failing closed.

Constraints:

- Database-level permissions are the primary control. The parser is defence-in-depth, not the guard. Assume the parser will eventually be bypassed and make sure that is survivable.
- Do not weaken any existing check to make tests pass.

Acceptance:

- All new tests pass.
- A prompt asking for user emails or password hashes returns a refusal or an authorisation error, never data - verify by running the chat path live.

P1 — Pillar registry and scope cut

Context: Navuuna currently renders 8 pillars, several with no real data source. We are narrowing to 3. See CLAUDE.md at the repo root for the scope.

Goal: make the codebase structurally incapable of rendering a pillar we cannot source, rather than relying on people remembering not to.

Tasks:

- Create a single pillar registry (one module, imported by backend, ingestion and frontend - or generated from one JSON if cross-language sharing is awkward). Each pillar declares:
 - key, display_name, status (active | held | off),
 - method (measured | proxy | gap),
 - granularity (subcounty | county | utility | national),
 - source_id, vintage, weight
- Populate it:
 - ACTIVE: water_sanitation (measured, subcounty)
 - road_density (measured, subcounty)
 - transit_access (measured, subcounty)
 - HELD: electricity_access (measured, subcounty, vintage "2019 census")

OFF: freedom_index, safety, project_momentum, environment, smart_grid

- Anything with status "off" must not be served by the API, must not appear in the frontend, and must not have a code path that could render it. Delete the dead components rather than hiding them behind a flag.
- Rename in UI and API: "Smart Grid" -> "Electricity Access", "Road Progress" -> "Road Density". Update all copy, keys and seeded data.
- Add a test that fails if a pillar with status "off" appears in any API response.

Constraints:

- Do not delete historical migrations. Add new ones.
- Keep the existing Mapbox map, styling, colour bands and side panel exactly as they are. This task changes what data exists, not how it is drawn.

P2 — Data package and the provenance contract

Context: Creating a new package `nuvola-atlas-data` that produces the data files the app consumes. It runs offline and is decoupled from the app by one contract: the shape of nairobi_vitality.geojson.

Tasks:

- Scaffold nuvola-atlas-data with the layout in the Refocus Workflow doc, section 5. Python 3.13, pinned deps, ruff + pytest.
- Define the provenance schema. Every indicator value is:


```
{
  "value": number | null,
  "unit": string,
  "indicator": string,          // from the canonical vocabulary
  "granularity": "subcounty" | "county" | "utility" | "national",
  "method": "measured" | "proxy" | "gap",
  "source_id": string,         // maps to a manifest
  "vintage": string,           // "FY2023/24", "2019 census"
  "retrieved": "YYYY-MM-DD",
  "page_ref": string | null    // for PDF-extracted values
}
```
- A manifest format (YAML, one per source) recording: name, url, licence, attribution string, vintage, retrieval date, sha256 of the raw file.
- Implement pipeline/boundaries.py: download COD-AB admin2, filter to Nairobi, write nairobi_subcounties.gpkg. Assert exactly 17 features or fail loudly.
- Implement pipeline/emit.py writing nairobi_vitality.geojson.

CRITICAL RULE - encode this in the emitter, not in documentation:

- method == "gap" -> value MUST be null. Never a number.
- granularity != "subcounty" -> the value goes in a top-level `county\_context` object, NEVER into a feature's properties.

Add tests asserting both. These tests are the product's integrity guarantee.

- Before finalising field names inside `properties`, read the current frontend and match the existing dummy-object keys exactly so the file drops in with no renaming. Report the names you found.

Constraints:

- No network calls at import time. Downloads are explicit pipeline steps.
- Every output file gets a companion entry in provenance.json.

P3 — The WASREB IMPACT parser

Context: WASREB (Kenya's water regulator) publishes an annual IMPACT report. 17 editions exist as PDFs at wasreb.go.ke/impact-reports. They contain utility-level performance data for up to 92 water service providers across 47 counties: non-revenue water, hours of supply, metering ratio, revenue collection efficiency, O&M cost coverage, water quality, staff productivity, coverage.

Nobody has turned this into a machine-readable time series. That series is our core asset. Accuracy matters more than coverage - a wrong number is worse than a missing one.

Tasks:

- Build pipeline/wasreb/ with:
 - a downloader that fetches each edition and records sha256 in a manifest
 - PER-EDITION extractors (table layouts and indicator sets change between years - do not attempt one universal parser)
 - a normaliser mapping each edition's column headers to a canonical

- indicator vocabulary defined in one place
- a utility-name resolver handling renames, mergers and spelling variants across years, with an explicit alias table a human maintains
- Output wasreb_impact_long.csv (and .parquet) with columns: utility_id, utility_name, county, fy, indicator, value, unit, report_issue, page_ref, extraction_confidence
- extraction_confidence: "high" (clean structured table), "medium" (reflowed or OCR'd), "low" (ambiguous). Never silently drop a low-confidence value - emit it flagged.
- Build a validation report: per edition, how many utilities and indicators were extracted, which pages failed, and any value outside a plausible range (e.g. a percentage above 100).
- Write a manual spot-check harness: sample 30 random values, print each with its page_ref, and record a human verdict in a checked-in file.

Constraints:

- Use pdfplumber for structured tables. Only fall back to OCR where genuinely needed, and mark those values medium confidence.
- Do not interpolate, estimate or forward-fill missing years. Missing is missing.
- Do not compute any sub-county figure from this data. It is utility-level and must be tagged granularity: "utility".

Acceptance:

- At least 10 editions parsed with a validation report per edition.
- The spot-check file exists with 30 human-verified values.

P4 — Join, score, and emit the map file

Context: Combining the boundaries, KNBS census tables and derived road/transit metrics into the single GeoJSON the Mapbox map reads.

Tasks:

- pipeline/knbs.py: load the 2019 census tables (population/density/households; % households by main water source; by toilet type; by lighting fuel), join to the 17 sub-counties by name with an explicit alias table for name mismatches. Fail loudly on any unmatched sub-county - never drop one silently.
- pipeline/osm_roads.py: clip HOT OSM Kenya roads to each sub-county, sum length, divide by area -> road density (km/km2).
- pipeline/gtfs_transit.py: load Digital Matatus GTFS, compute per sub-county the share of population within 500 m of a stop, using WorldPop for the population surface. Document the buffer distance as a parameter, not a magic number.
- pipeline/score.py: scale each indicator to 0-100 across the 17 sub-counties. All weights in ONE versioned dict with a comment explaining each. Emit the weights into provenance.json so the index is defensible.
- pipeline/emit.py: write nairobi_vitality.geojson with the agreed property names, plus a top-level county_context object carrying the utility-level WASREB values for Nairobi (NCWSC).

Constraints:

- Re-read the CRITICAL RULE from P2. The county_context separation is the point of this whole exercise.
- OSM coverage is biased: wealthier areas are mapped more completely than informal settlements. Record this as a caveat in provenance.json against every OSM-derived indicator so the frontend can surface it.
- The pipeline must be idempotent and rerunnable from manifests alone.

P5 — Frontend: make provenance visible

Context: Our differentiator is that we never present an unsourced number. Right now that discipline lives in a document. Make it a visible product feature.

Tasks:

- Swap the hardcoded dummy sub-county array for a fetch of /data/nairobi_vitality.geojson. Keep the Mapbox setup, bubble styling, colour bands and side panel exactly as they are.
- Every indicator shown in the side panel displays its source and vintage as small secondary text - e.g. "KNBS 2019 Census" or "WASREB IMPACT 17 . FY2023/24".
- Pillars with method "gap": render greyed with "Limited data" and no number. They must be visually distinct at a glance, not hidden.
- Render county_context in a banner ABOVE the map, clearly labelled as county-wide (e.g. "Nairobi County . Non-revenue water 51% . NCWSC . FY2023/24"). It must be visually impossible to mistake for a sub-county value.

- Add a "Data sources" panel listing every dataset with its licence and attribution. OpenStreetMap and WorldPop attribution is licence-required.
- Where an indicator carries an OSM coverage caveat, show an info affordance explaining that under-mapped areas can appear to have fewer services.

Constraints:

- No new dependencies unless genuinely necessary.
- Do not change the colour thresholds or bubble geometry.

P6 — Performance and reliability (from the audit)

Context: The audit flagged scalability issues. Lower priority than P0 but they will bite as data grows.

Tasks:

- IngestController: readings are applied row-by-row inside a transaction, causing repeated updates to the same zones row and lock contention that blocks concurrent API reads. Group readings by zone_id in memory first, then issue ONE update per zone.
- Add PostGIS GIST spatial indexes on all geography/geometry columns (zones.centroid and any boundary columns). Add via migration and verify with EXPLAIN that the relevant queries use them.
- FastAPI: batches of up to 5,000 entries are validated synchronously with Pydantic, blocking the event loop and timing out /api/health. Move heavy list validation to asyncio.to_thread.
- The FastAPI -> Laravel signature check allows 300s clock skew. Add a healthcheck that alerts on clock drift between the two hosts, and document NTP as a deployment requirement.

Acceptance:

- A benchmark showing ingestion of 5,000 readings does not block /api/health.
- EXPLAIN output confirming index usage on a representative spatial query.

SECTION 8

Institutional workflow — Track C

Runs in parallel with everything above. Blocks nothing.

WHEN	ACTION	TO	ANCHOR TO
Week 1	Letter requesting participation	Kenya Space Agency	The EO Data Sharing Framework 2026 and the urban-observatory student-engagement item. Strathmore and Daystar on the letterhead.
Week 1	Register accounts	tenders.go.ke · WBGeProcure	So K-WASH terms of reference and expressions of interest reach you directly.
Week 2–3	Meeting request	Nairobi City County	Their OGP Action Plan 2025 – June 2027 Open Data Portal commitment.
Week 4	Introduction + the published series	WASREB	"We turned your 17 IMPACT reports into a queryable series. Attributed to you. Free."
Month 3+	Grant application	Lacuna Fund	With the WASREB series as evidence of execution.

SEQUENCING RULE

Never approach WASREB or the county before the series exists. The whole strategy rests on arriving having already done the work — not asking permission to start it.

SECTION 9

Definition of done for this phase

- ☐ P0 merged; a chat prompt asking for user emails returns an error, verified live
- ☐ Codebase renders exactly 3 pillars; "off" pillars have no code path at all
- ☐ wasreb_impact_long.csv covers at least 10 editions with a validation report
- ☐ 30 values hand-verified against the source PDFs, recorded in the repo
- ☐ nairobi_vitality.geojson drops into the map with zero frontend renaming
- ☐ Every number on screen shows its source and vintage
- ☐ Gap pillars render grey with no number
- ☐ County-level values appear only in the banner, never on a bubble
- ☐ Three letters sent

SECTION 10

What "off" means

Off means deleted, not flagged. A feature flag is an invitation to turn something back on at 2am before a demo. If Freedom Index, Safety or Project Momentum returns, it returns through a written decision recorded in this document — with a named data source that actually exists.

The failure mode for this project is not a bad decision. It is a good decision that quietly erodes.